

MODULE 40 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Hybrid and Coexistence

Closing Domain 4. The transition period is an architecture, not an inconvenience.

DOMAIN 4 of 4 EXAM WEIGHT 20% MODULE 40 of 44

The hybrid period is not a gap between two architectures. **It is one, and it will outlive your estimate.**

Three things must span both estates

- ◆ **Network.** Direct Connect for predictable bandwidth, with VPN as backup, and Transit Gateway as the hub. Modules six and seven.
- ◆ **Identity.** One directory both sides trust, so a person has one account. AD trusts or Managed AD, from module four.
- ◆ **DNS.** Resolution must work in both directions, which is Route 53 Resolver inbound and outbound endpoints. Module eight.
- ◆ Get any of these wrong and every wave afterwards inherits the problem, because every wave depends on all three.

ON THE EXAM

These are foundation work, done once, before wave one. Retrofitting identity across a half-migrated estate is considerably worse.

Data gravity, and the chatty dependency

- ◆ **Split a chatty pair across estates and every call now crosses the link**, adding latency and cost to each one.
- ◆ **Data has gravity.** Applications want to be near their data, so moving the database usually decides the wave.
- ◆ Dependency mapping from module 33 is what tells you which pairs are chatty. Guessing produces the problem.
- ◆ A latency-sensitive dependency may add milliseconds per call. At thousands of calls per transaction, that is seconds.

ON THE EXAM

The application that worked fine on-premises and is slow in AWS is usually still calling something on-premises, thousands of times.

What a long hybrid costs

- ◆ **Two of everything.** Two monitoring stacks, two identity paths, two runbooks, two on-call rotas.
- ◆ **Data transfer across the link**, charged per gigabyte, for every conversation you split. Module twenty two.
- ◆ **The data centre does not get cheaper as it empties.** Rent, power and support contracts are largely fixed.
- ◆ Attention is the hidden cost. Teams running two estates are not improving either of them.

ON THE EXAM

A half-empty data centre costs nearly what a full one does. That is why finishing matters more than starting well.

When something genuinely cannot move

- ◆ **AWS Outposts** puts AWS infrastructure and APIs in your own facility, for workloads with local dependencies or residency requirements.
- ◆ AWS names **migration of applications with local system interdependencies** as an Outposts use case directly.
- ◆ Second-generation racks run C7i, M7i and R7i, roughly 40% faster than first-generation. Servers cover branch and retail sites.
- ◆ It needs real facilities: floor loading, 10 kVA power, cooling, and a service link of at least 500 Mbps to the parent Region.

ON THE EXAM

Outposts is for the workload that cannot leave, not for the workload nobody wants to move. Those are very different problems.

One operational model

- ◆ **Systems Manager hybrid activations** make on-premises servers managed nodes, so patching, inventory and access are identical. Module 26.
- ◆ **One observability stack**, with on-premises telemetry flowing to CloudWatch, so an incident has one timeline.
- ◆ One set of runbooks, one on-call rota, one place to look. The alternative is two of each for years.
- ◆ This is the strongest practical argument for hybrid tooling, and it is usually made too late to help.

ON THE EXAM

An incident spanning both estates with two toolsets means two timelines and no correlation. That is module 23's problem, doubled.

Reading the scenario

THE SCENARIO SAYS

An app is slow after migration, calls on-premises

Users need accounts in both environments

On-premises cannot resolve AWS private names

Must sit beside factory equipment

Two monitoring stacks, two on-call rotas

The hybrid has run for three years

IT IS POINTING AT

A chatty dependency split across estates

One directory, with a trust

Route 53 Resolver outbound endpoint

AWS Outposts

One operational model, hybrid activations

Finish it. Half-migrated is the worst state

ON THE EXAM

Most coexistence problems are one of three: something split that should not be, something that cannot span, or something never finished.

Meridian Freight migrated its quoting service to AWS in wave two. It now takes 4 seconds to produce a quote, against 300 milliseconds before. The service makes roughly 1,200 calls to a pricing database that remains on-premises, per quote.

The Direct Connect link adds about 3 milliseconds each way.

What should the solutions architect recommend?

- ☐ A Migrate the pricing database into the same wave as the quoting service, keeping the chatty pair together
- ☐ B Upgrade the Direct Connect link to a higher bandwidth
- ☐ C Add a second Direct Connect connection for redundancy
- ☐ D Increase the quoting service's instance size

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Move the database with it
1,200 calls at 6 ms round trip is about 7 seconds of pure latency. Only removing the crossing fixes it, which means the pair moves together.
- B** More bandwidth
Bandwidth is throughput, not latency. A wider link does not shorten the distance the packets travel.
- C** Second connection
Resilience, not speed. Two links of the same latency give the same round trip.
- D** Bigger instance
The service is waiting on the network, not short of CPU. A larger instance waits at the same speed.

THE SIGNAL

The arithmetic is given to you: 1,200 calls times 6 ms. Bandwidth and latency are different properties, and only one is the problem.

During a three-year migration, staff will need to access applications in both the data centre and AWS. The proposal is to create separate AWS-native accounts for every user, with a manual process to keep passwords aligned.

There are 6,000 staff and an existing on-premises Active Directory.

What should the solutions architect recommend?

- A Extend the existing directory: AWS Managed Microsoft AD with a trust to on-premises, so each person keeps one identity
- B Create the separate accounts as proposed, with password synchronisation
- C Wait until the migration completes, then build identity in AWS
- D Give every user an IAM user with an access key

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Managed AD with a trust
One identity per person across both estates, with existing groups and policies intact. Joiners and leavers stay a single process.
- B** Separate accounts, synced
12,000 identities and a manual sync for three years. A leaver disabled in one directory keeps access through the other.
- C** Wait until the end
Staff need access to AWS applications from wave one. Deferring identity means every wave invents its own arrangement.
- D** IAM users with keys
Long-lived credentials for 6,000 people, which module 28 spent a module removing. It also does not authenticate to on-premises applications.

THE SIGNAL

Two identities per person means a leaver can be disabled in one and still have access through the other. That is the security failure.

After wave one, on-premises applications cannot resolve the private DNS names of migrated services, and migrated services cannot resolve on-premises hostnames. Teams have been adding entries to hosts files as a workaround, and several are now stale.

There is a Direct Connect link and a Route 53 private hosted zone.

What should the solutions architect implement?

- ☐ A Route 53 Resolver inbound and outbound endpoints with forwarding rules in both directions
- ☐ B A script that distributes an updated hosts file to all servers nightly
- ☐ C Move all DNS to the on-premises servers
- ☐ D Use public DNS names for the migrated services

◆ PAUSE AND DECIDE

What should the solutions architect implement?

- A** Inbound and outbound endpoints
Inbound lets on-premises query AWS; outbound lets AWS query on-premises. The problem is bidirectional, so both are needed.
- B** Distributed hosts files
Automates the workaround. Hosts files bypass DNS entirely, so every IP change becomes a fleet-wide update that is stale by definition.
- C** All DNS on-premises
On-premises servers cannot resolve Route 53 private hosted zones natively. It also makes the data centre a dependency you are trying to remove.
- D** Public DNS names
Exposes internal service names publicly, and private IP addresses in public DNS do not resolve usefully anyway.

THE SIGNAL

Module eight's distinction: inbound means on-premises asks AWS, outbound means AWS asks on-premises. Both directions fail here, so both endpoints.

A manufacturer must close its data centre, but one application controls production line equipment over a protocol requiring sub-millisecond response, physically wired within the plant. Everything else can migrate.

The plant has a suitable room, power and cooling, and a 1 Gbps connection.

What should the solutions architect recommend?

- ☐ A AWS Outposts in the plant for the control application, migrating everything else to the Region
- ☐ B Migrate the control application and accept the latency over Direct Connect
- ☐ C Retain a small data centre for the control application indefinitely
- ☐ D Use a Local Zone for the control application

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Outposts in the plant
AWS infrastructure physically beside the equipment, with the same APIs and operational model as the Region. The facility requirements are met.
- B** Accept the latency
Sub-millisecond over any wide-area link is physically impossible. The requirement is not a preference to be negotiated.
- C** Keep a small data centre
Meets the latency need and keeps a facility, a support contract and a second operational model for one application.
- D** Local Zone
Local Zones sit in metropolitan areas, not in your plant. Closer than a Region, and still not on the factory floor.

THE SIGNAL

Outposts is for what cannot leave. Sub-millisecond to wired equipment is a physical constraint, not an architectural preference.

An incident spanning both estates took six hours to diagnose. On-premises servers are monitored by a legacy tool, AWS resources by CloudWatch, and correlating the two timelines was done manually in a spreadsheet. Patching and access follow two different processes. The hybrid period has two more years to run.

What should the solutions architect recommend?

- A Register on-premises servers as managed nodes with SSM hybrid activations, and send their telemetry to CloudWatch for one timeline
- B Deploy the legacy monitoring tool into AWS as well
- C Accept two toolsets, since the hybrid is temporary
- D Accelerate the migration so the hybrid ends sooner

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Hybrid activations plus one timeline
One operational model across both estates: the same patching, inventory and session access, and a single correlated incident timeline.
- B

 Legacy tool into AWS
Defensible, and it means licensing and running that tool across a growing estate while losing AWS-native integration.
- C

 Accept two toolsets
Two more years is not temporary. It is longer than most teams keep the same tooling voluntarily.
- D

 Accelerate the migration
Changes the programme timeline to avoid an operational problem that has a direct fix.

THE SIGNAL

Two more years is the planted detail. It converts "temporary inconvenience" into a design decision that needs making.

A migration begun three years ago is 70% complete and has stalled. The remaining 30% are applications nobody wants to own. The data centre still costs \$2.1m a year, barely less than when it was full, and two operational models are still being run. Leadership has asked whether to simply stop where things are.

What should the solutions architect advise?

- ☐ A Finish it: assign the remaining 30% to the seven Rs, retiring and repurchasing aggressively, because half-migrated is the most expensive state
- ☐ B Stop where things are and run permanently hybrid
- ☐ C Migrate the remaining 30% as-is with no assessment, to finish quickly
- ☐ D Move the migrated 70% back on-premises to consolidate

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A** Finish, with aggressive retire and repurchase
The data centre cost is fixed until it closes, so only finishing removes it. Applications nobody wants to own are prime retire and repurchase candidates.
- B** Stop and stay hybrid
Locks in \$2.1m a year and two operational models permanently, to avoid finishing 30% of the work.
- C** Migrate as-is, no assessment
Fast, and it migrates applications that should be retired or repurchased. Speed without assessment moves the waste rather than removing it.
- D** Move back on-premises
Discards three years of work and re-provisions capacity in a facility the business wants to exit.

THE SIGNAL

The data centre bill barely fell at 70% migrated, which is the whole argument. The saving arrives only when the doors close.

What to carry out of this module

- ◆ **Design the transition deliberately.** Network, identity and DNS must span both estates before wave one.
- ◆ **Never split a chatty pair.** Bandwidth is not latency, and a wider link does not shorten the distance.
- ◆ **Run one operational model,** using hybrid activations and one observability stack. Two of everything for years is the real cost.
- ◆ **A half-empty data centre costs nearly what a full one does.** The saving arrives when it closes, not before.

ON THE EXAM

Outposts is for what cannot leave. Applications nobody wants to own are usually retire or repurchase candidates, not migration ones.